

Rwanda Legal AI — Ready-to-Deploy on Render (Full Project)

Iyi version ni **Render-ready**, ikubiyemo backend, frontend, auth, MTN MoMo payments, HTTPS automatic, na CI/CD. Hano twahinduye project structure n'amafayili kugira ngo ushobore kuyideploy ako kanya.

Project Structure

```
rwanda-legal-ai/
├── backend/
│   ├── app/
│   ├── requirements.txt
│   ├── Procfile # Flask/Django start command
│   └── .env      # Optional for local testing
├── frontend/
│   ├── src/
│   ├── package.json
│   ├── .env      # Optional for local testing
│   └── build/    # auto-generated after build
├── data/
├── nginx.conf
├── docker-compose.yml
├── .env.example
├── README.md
└── .github/workflows/deploy.yml
```

Backend — Procfile

Flask:

```
web: gunicorn app:app
```

Django:

```
web: gunicorn backend.wsgi
```

Render izakoresha Procfile kugira ngo itangize backend.

Frontend — package.json scripts

```
"scripts": {  
  "start": "react-scripts start",  
  "build": "react-scripts build",  
  "serve": "serve -s build"  
}
```

- Render izakoresha `npm run build` kugira ngo ikore production build. - Publish directory: `build`

Render Deployment Steps

1. **Fork/Clone repo** kuri GitHub.
 2. **Backend (Web Service):**
 3. Environment: Python 3.10+
 4. Build command: `pip install -r requirements.txt`
 5. Start command: `web: gunicorn app:app` (Flask) cyangwa `gunicorn backend.wsgi` (Django)
 6. Env variables: `SECRET_KEY`, `DATABASE_URL`, `MTN_API_KEY`, `MOMO_USER_ID`, etc.
 7. **Database:** PostgreSQL free tier muri Render (shyira `DATABASE_URL` mu env variables).
 8. **Frontend (Static Site):**
 9. Build command: `npm install && npm run build`
 10. Publish directory: `build`
 11. Env variable: `REACT_APP_API_URL=https://<your-backend>.onrender.com`
 12. Render izaguha HTTPS free automatic.
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MTN MoMo Setup (sandbox)

- Shyira MTN sandbox credentials muri env variables.
 - Backend route: `POST /api/payments/subscribe` izajya icunga subscription.
 - Uzashobora gusuzuma payments mbere yo kwimura kuri production keys.
-

CI/CD (GitHub Actions)

- Workflow: `.github/workflows/deploy.yml`
- Trigger: Push kuri `main` branch
- Steps: Checkout repo → SSH kuri server (optional if using VPS) → docker-compose rebuild → run migrations (optional)

- Kubaka kuri Render, Render iracyakora auto-build kuri push kuri main branch.
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Ubu ushobora:

1. Kugira **backend na frontend ready-to-deploy** ku Render.
2. Kugenzura app na sandbox MTN MoMo payments.
3. Kuzirikana CI/CD kuri GitHub Actions igihe uzajya udeploy kuri VPS cyangwa production server.

Iyi project ubu iri **Render-ready**, ushobora gukurikiza steps zo hejuru ugatangira deployment ako kanya.